

Ideal_LTSpice_Netlist_4

Prompt

What should be modified in the netlist, and why ? Give the modified version of the netlist, precising and justifying every modification made.

Answer

To get your AM radio working, we need to apply a two-pronged fix to overcome the "brick wall" at the detector stage. We will replace the standard diode with a Germanium equivalent and boost the RF amplifier's gain to ensure the signal is strong enough to be detected.

Here is the modified netlist, followed by the exact justifications for the changes.

```
* AM Radio Receiver - Corrected for Diode Threshold
* -----
* POWER SUPPLY
V1 VCC 0 9V

* ANTENNA & AM SOURCE (Tuned to BBC 5 Live: 693 kHz, 1 kHz Tone)
B_ANT ANT 0 V=0.05 * (1 + 0.8*sin(2*pi*1k*time)) * sin(2*pi*693k*time)
R_ANT ANT TANK_IN 50

* LC TUNING STAGE (693 kHz)
L_TUNE TANK_IN 0 200u
C_TUNE TANK_IN 0 264p

* RF AMPLIFIER (NPN)
C_RF_IN TANK_IN RF_B 10n
Q_RF RF_C RF_B RF_E NPN_IDEAL
R_RF_B VCC RF_B 1Meg
R_RF_C VCC RF_C 4.7k
R_RF_E RF_E 0 1k
C_RF_BYPASS RF_E 0 100n ; <--- MODIFICATION 1: Emitter Bypass

C_RF_OUT RF_C DET_IN 10n

* ENVELOPE DETECTOR (Demodulator)
D_DET DET_IN DET_OUT 1N34A ; <--- MODIFICATION 2: Germanium Diode
R_DET DET_OUT 0 10k
C_DET DET_OUT 0 10n

* AUDIO AMPLIFIER (NPN)
```

```

C_AUD_IN DET_OUT AUD_B 10u
Q_AUD AUD_C AUD_B AUD_E NPN_IDEAL
R_AUD_B VCC AUD_B 1Meg
R_AUD_C VCC AUD_C 4.7k
R_AUD_E AUD_E 0 1k
C_AUD_OUT AUD_C OUT 10u

* LOAD (Headphones)
R_LOAD OUT 0 32

* MODELS
.model NPN_IDEAL NPN(Bf=100)
.model 1N34A D(Is=200n Rs=2 N=1.2 Cjo=1p Eg=0.67) ; <--- MODIFICATION 2:
Germanium Parameters

* SIMULATION COMMAND
.tran 0 5m 0 0.1u
.end

```

Detailed Justifications for Modifications

1. Adding an RF Emitter Bypass Capacitor (C_RF_BYPASS)

- **What was changed:** Added `C_RF_BYPASS RF_E 0 100n` in parallel with the RF amplifier's emitter resistor (`R_RF_E`).
- **Why it was changed:** In your original circuit, the voltage gain of the RF transistor was heavily restricted by the 1 k Ω emitter resistor to roughly a factor of 4.7 (calculated as Collector Resistor / Emitter Resistor). By placing a 100 nF capacitor across the emitter resistor, the DC bias remains perfectly stable, but we create a near-zero resistance path to ground specifically for the 693 kHz high-frequency AC signal. This unlocks the maximum potential AC gain of the transistor, driving a significantly larger signal into the detector to guarantee it triggers.

2. Switching to a Germanium Diode Model (1N34A)

- **What was changed:** Replaced `D_IDEAL` with `1N34A` and added its corresponding SPICE `.model` parameters (`Is=200n Rs=2 N=1.2 Cjo=1p Eg=0.67`).
- **Why it was changed:** The previous ideal diode model defaulted to standard Silicon behavior, requiring a ~600 mV forward voltage drop to conduct. Your original signal was only swinging ± 440 mV, so the diode blocked it entirely. Germanium diodes, such as the classic 1N34A used in nearly all vintage AM radios, have a much lower forward voltage

requirement (around 200 mV to 300 mV). Combined with the boosted RF gain from modification 1, this ensures the AC signal effortlessly clears the threshold to successfully rectify the audio envelope.

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